

Assignment 5 - Linear Regression

AI & ML Summer Internship 2026

Name: Abhiney Kumar

Enrollment Number: 198409

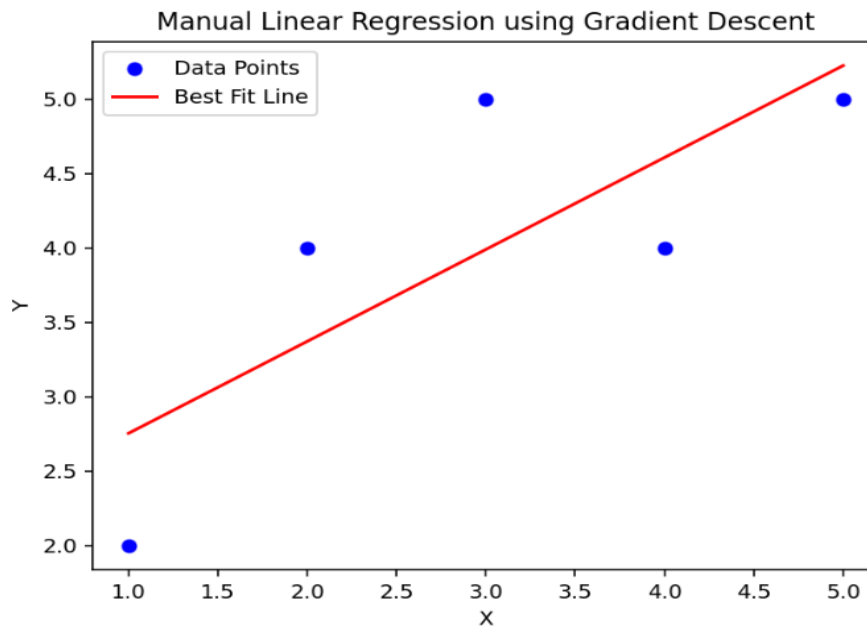
Department: Electronics and Communication Engineering (ECE)

Institute: NIT Jalandhar

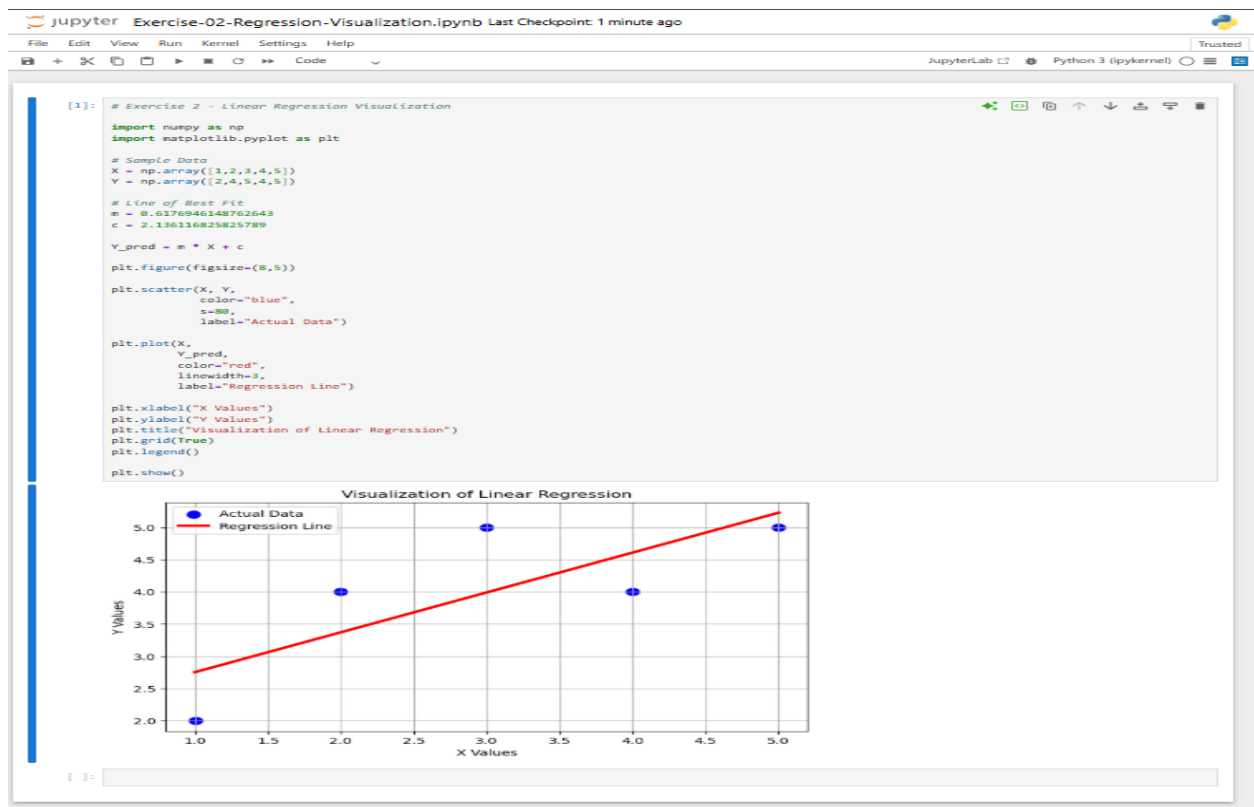
Objective

Implement Linear Regression manually using Gradient Descent, visualize the regression line, and compare the results with Scikit-Learn.

Exercise 1 - Manual Gradient Descent



Exercise 2 - Regression Visualization



Exercise 3 - Scikit-Learn Code

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[1]: # Exercise 3 - Linear Regression using Scikit-Learn

import numpy as np
import matplotlib.pyplot as plt
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score

# Sample Dataset
X = np.array([1,2,3,4,5]).reshape(-1,1)
Y = np.array([2,4,5,4,5])

# Create Model
model = LinearRegression()

# Train Model
model.fit(X, Y)

# Predictions
Y_pred = model.predict(X)

# Results
print("Slope (Coefficient):", model.coef_[0])
print("Intercept:", model.intercept_)
print("Mean Squared Error:", mean_squared_error(Y, Y_pred))
print("R² Score:", r2_score(Y, Y_pred))

# Predict new value
new_x = np.array([[6]])
prediction = model.predict(new_x)

print("Prediction for X = 6 :", prediction[0])

# Plot
plt.figure(figsize=(8,5))

plt.scatter(X, Y,
            color="blue",
            s=80,
            label="Actual Data")

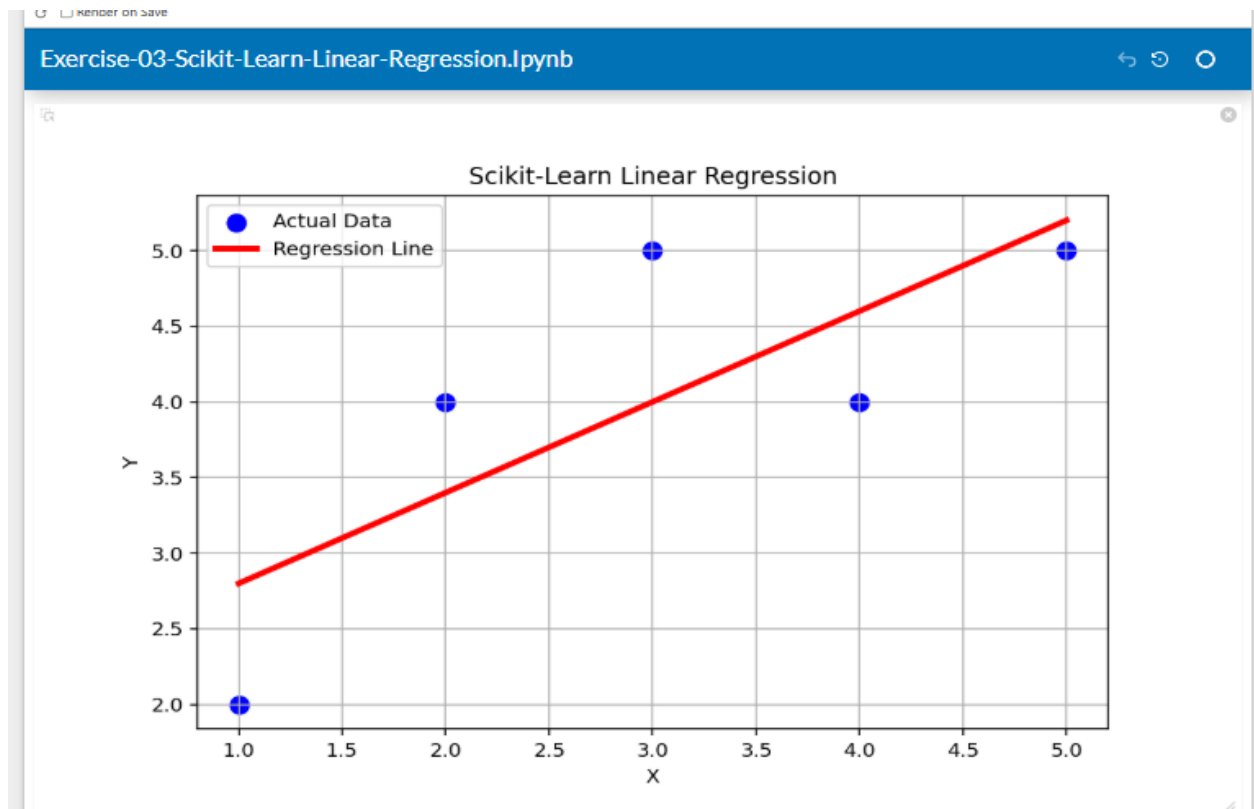
plt.plot(X,
         Y_pred,
         color="red",
         linewidth=3,
         label="Regression Line")

plt.xlabel("X")
plt.ylabel("Y")
plt.title("Scikit-Learn Linear Regression")
plt.grid(True)
plt.legend()

plt.show()
```

Slope (Coefficient): 0.6000000000000002
Intercept: 2.1999999999999993
Mean Squared Error: 0.4799999999999998
R² Score: 0.6000000000000001
Prediction for X = 6 : 5.800000000000001

Exercise 3 - Scikit-Learn Graph



Conclusion

This assignment demonstrated Linear Regression through manual Gradient Descent and the Scikit-Learn library. Both implementations successfully generated a regression model and visualization, reinforcing the fundamentals of supervised machine learning.